Neural Computation -- 1997

103 article(s), all issues

1. A Local Learning Rule That Enables Information Maximization for Arbitrary Input Distributions

LinskerRalph

https://doi.org/10.5555/1246443.1246445

This note presents a local learning rule that enables a network to maximize the mutual information between input and output vectors. The network's output units may be nonlinear, and the distributio...

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2. Adaptive Mixtures of Probabilistic Transducers

SingerYoram

https://doi.org/10.5555/1246443.1246449

We describe and analyze a mixture model for supervised learning of probabilistic transducers. We devise an online learning algorithm that efficiently infers the structure and estimates the paramete...

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3. Minimax Entropy Principle and Its Application to Texture Modeling

ZhuSong Chun, WuYing Nian, MumfordDavid

https://doi.org/10.5555/1246443.1246444

This article proposes a general theory and methodology, called the minimax entropy principle, for building statistical models for images (or signals) in a variety of applications. This principle co...

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4. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1997.9.issue-1

No abstract available.

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5. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1997.9.issue-7

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6. (untitled)

(authors unknown)

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7. (untitled)

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(authors unknown)

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9. (untitled)

(authors unknown)

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10. (untitled)

(authors unknown)

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No abstract available.

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11. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1997.9.issue-2

No abstract available.

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12. Flat Minima

Sepp Hochreiter, Jürgen Schmidhuber

https://doi.org/10.1162/neco.1997.9.1.1 | pp. 1-42

We present a new algorithm for finding low-complexity neural networks with high generalization capability. The algorithm searches for a "flat" minimum of the error function. A flat minimum is a large connected region

in weight space where the error remains approximately constant. An MDL-based, Bayesian argument suggests that flat minima correspond to "simple" networks and low expected overfitting. The argument is based on a Gibbs algorithm variant and a novel way of splitting generalization error into underfitting and overfitting error. Unlike many previous approaches, ours does not require gaussian assumptions and does not depend on a "good" weight prior. Instead we have a prior over input-output functions, thus taking into account net architecture and training set. Although our algorithm requires the computation of second-order derivatives, it has backpropagation's order of complexity. Automatically, it effectively prunes units, weights, and input lines. Various experiments with feedforward and recurrent nets are described. In an application to stock market prediction, flat minimum search outperforms conventional backprop, weight decay, and "optimal brain surgeon/optimal brain damage".

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13. Lyapunov Functions for Neural Nets with Nondifferentiable Input-Output Characteristics

Jianfeng Feng

https://doi.org/10.1162/neco.1997.9.1.43 | pp. 43-49

I construct Lyapunov functions for asynchronous dynamics and synch ronous dynamics of neural networks with nondifferentiable input-output characteristics.

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14. Detecting Synchronous Cell Assemblies with Limited Data and Overlapping Assemblies

Gary Strangman

https://doi.org/10.1162/neco.1997.9.1.51 | pp. 51-76

Two statistical methods-cross-correlation (Moore et al. 1966) and gravity clustering (Gerstein et al. 1985)-were evaluated for their ability to detect synchronous cell assemblies from simulated spike train data. The two methods were first analyzed for their temporal sensitivity to synchronous cell assemblies. The presented approach places a lower bound on the amount of data required to detect a synchronous assembly. On average, both methods required the same minimum amount of recording time to detect significant pairwise correlations, but the gravity method exhibited less variance in the recording time. The precise length of recording depends on the consistency with which a neuron fires synchronously with the assembly but was independent of the assembly firing rate. Next, the statistical methods were tested with respect to their ability to differentiate two distinct assemblies that overlapped in time and space. Both statistics could adequately differentiate two overlapping synchronous assemblies. For cross-correlation, this ability deteriorates quickly when considering three or more simultaneously active, overlapping assemblies, whereas the gravity method should be more flexible in this regard. The work demonstrates the difficulty of detecting assembly phenomena from simultaneous neuronal recordings. Other statistical methods and the detection of other types of assemblies are also discussed.

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15. A Simple Neural Network Exhibiting Selective Activation of Neuronal Ensembles: From Winner-Take-All to Winners-Share-All

Tomoki Fukai, Shigeru Tanaka
https://doi.org/10.1162/neco.1997.9.1.77 | pp. 77-97

A neuroecological equation of the Lotka-Volterra type for mean firing rate is derived from the conventional membrane dynamics of a neural network with lateral inhibition and self-inhibition. Neural selection mechanisms employed by the competitive neural network receiving external inputs are studied with analytic and numerical calculations. A remarkable findings is that the strength of lateral inhibition relative to that of self-inhibition is crucial for determining the steady states of the network among three qualitatively different types of behavior. Equal strength of both types of inhibitory connections leads the network to the well-known winner-take-all behavior. If, however, the lateral inhibition is weaker than the self-inhibition, a certain number of neurons are activated in the steady states or the number of winners is in general more than one (the winners-share-all behavior). On the other hand, if the self-inhibition is weaker than the lateral one, only one neuron is activated, but the winner is not necessarily the neuron receiving the largest input. It is suggested that our simple network model provides a mathematical basis for understanding neural selection mechanisms.

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16. Playing Billiards in Version Space

P. Ruján
https://doi.org/10.1162/neco.1997.9.1.99 | pp. 99-122

A ray-tracing method inspired by ergodic billiards is used to estimate the theoretically best decision rule for a given set of linear separable examples. For randomly distributed examples, the billiard estimate of the single Perceptron with best average generalization probability agrees with known analytic results, while for real-life classification problems, the generalization probability is consistently enhanced when compared to the maximal stability Perceptron.

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17. Partial BFGS Update and Efficient Step-Length Calculation for Three-Layer Neural Networks

Kazumi Saito, Ryohei Nakano
https://doi.org/10.1162/neco.1997.9.1.123 | pp. 123-141

Second-order learning algorithms based on quasi-Newton methods have two problems. First, standard quasi-Newton methods are impractical for large-scale problems because they require N2 storage space to maintain an approximation to an inverse Hessian matrix (N is the number of weights). Second, a line search to calculate a reasonably accurate step length is indispensable for these algorithms. In order to provide desirable performance, an efficient and reasonably accurate line search is needed. To overcome these problems, we propose a new second-order learning algorithm. Descent direction is calculated on the basis of a partial Broydon-Fletcher-Goldfarb-Shanno (BFGS) update with 2Ns memory space (s < < N), and a reasonably accurate step length is efficiently calculated as the minimal point of a second-order approximation to the objective function with respect to the step length. Our experiments, which use a parity problem and a speech synthesis problem, have shown that the proposed algorithm outperformed major learning algorithms. Moreover, it turned out that an efficient and accurate step-length calculation plays an important role for the convergence of quasi-Newton algorithms, and a partial BFGS update greatly saves

storage space without losing the convergence performance.

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18. Neural Networks for Functional Approximation and System Identification

H. N. Mhaskar, Nahmwoo Hahm

https://doi.org/10.1162/neco.1997.9.1.143 | pp. 143-159

We construct generalized translation networks to approximate uniformly a class of nonlinear, continuous functionals defined on Lp ([-1, 1]s) for integer s > or = 1, 1 < or = p < infinity, or C ([-1, 1]s). We obtain lower bounds on the possible order of approximation for such functionals in terms of any approximation process depending continuously on a given number of parameters. Our networks almost achieve this order of approximation in terms of the number of parameters (neurons) involved in the network. The training is simple and noniterative; in particular, we avoid any optimization such as that involved in the usual backpropagation.

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19. Selecting Optimal Experiments for Multiple Output Multilayer Perceptrons

Lisa M. Belue, Kenneth W. Bauer, Dennis W. Ruck

https://doi.org/10.1162/neco.1997.9.1.161 | pp. 161-183

Where should a researcher conduct experiments to provide training data for a multilayer perceptron? This question is investigated, and a statistical method for selecting optimal experimental design points for multiple output multilayer perceptrons is introduced. Multiple class discrimination problems are examined using a framework in which the multilayer perceptron is viewed as a multivariate nonlinear regression model. Following a Bayesian formulation for the case where the variance-covariance matrix of the responses is unknown, a selection criterion is developed. This criterion is based on the volume of the joint confidence ellipsoid for the weights in a multilayer perceptron. An example is used to demonstrate the superiority of optimally selected design points over randomly chosen points, as well as points chosen in a grid pattern. Simplification of the basic criterion is offered through the use of Hadamard matrices to produce uncorrelated outputs.

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20. A Penalty-Function Approach for Pruning Feedforward Neural Networks

Rudy Setiono

https://doi.org/10.1162/neco.1997.9.1.185 | pp. 185-204

This article proposes the use of a penalty function for pruning feedforward neural network by weight elimination. The penalty function proposed consists of two terms. The first term is to discourage the use of unnecessary connections, and the second term is to prevent the weights of the connections from taking excessively large values. Simple criteria for eliminating weights from the network are also given. The effectiveness of this penalty function is tested on three well-known problems: the contiguity problem, the parity problems, and the monks problems. The resulting pruned networks obtained for many of these problems have fewer connections than previously reported in the literature.

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21. Extracting Rules from Neural Networks by Pruning and Hidden-Unit Splitting

Rudy Setiono

https://doi.org/10.1162/neco.1997.9.1.205 | pp. 205-225

An algorithm for extracting rules from a standard three-layer feedforward neural network is proposed. The trained network is first pruned not only to remove redundant connections in the network but, more important, to detect the relevant inputs. The algorithm generates rules from the pruned network by considering only a small number of activation values at the hidden units. If the number of inputs connected to a hidden unit is sufficiently small, then rules that describe how each of its activation values is obtained can be readily generated. Otherwise the hidden unit will be split and treated as output units, with each output unit corresponding to an activation value. A hidden layer is inserted and a new subnetwork is formed, trained, and pruned. This process is repeated until every hidden unit in the network has a relatively small number of input units connected to it. Examples on how the proposed algorithm works are shown using real-world data arising from molecular biology and signal processing. Our results show that for these complex problems, the algorithm can extract reasonably compact rule sets that have high predictive accuracy rates.

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22. Probabilistic Independence Networks for Hidden Markov Probability Models

Padhraic Smyth, David Heckerman, Michael I. Jordan

https://doi.org/10.1162/neco.1997.9.2.227 | pp. 227-269

Graphical techniques for modeling the dependencies of random variables have been explored in a variety of different areas, including statistics, statistical physics, artificial intelligence, speech recognition, image processing, and genetics. Formalisms for manipulating these models have been developed relatively independently in these research communities. In this paper we explore hidden Markov models (HMMs) and related structures within the general framework of probabilistic independence networks (PINs). The paper presents a self-contained review of the basic principles of PINs. It is shown that the well-known forward-backward (F-B) and Viterbi algorithms for HMMs are special cases of more general inference algorithms for arbitrary PINs. Furthermore, the existence of inference and estimation algorithms for more general graphical models provides a set of analysis tools for HMM practitioners who wish to explore a richer class of HMM structures. Examples of relatively complex models to handle sensor fusion and coarticulation in speech recognition are introduced and treated within the graphical model framework to illustrate the advantages of the general approach.

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23. Using Expectation-Maximization for Reinforcement Learning

Peter Dayan, Geoffrey E. Hinton

https://doi.org/10.1162/neco.1997.9.2.271 | pp. 271-278

We discuss Hinton's (1989) relative payoff procedure (RPP), a static reinforcement learning algorithm whose foundation is not stochastic gradient ascent. We show circumstances under which applying the RPP is guaranteed to increase the mean return, even though it can make large changes in the values of the parameters. The proof is based on a mapping between the RPP and a form of the expectation-maximization

procedure of Dempster, Laird, and Rubin (1977).

24. Fast Sigmoidal Networks via Spiking Neurons

Wolfgang Maass

https://doi.org/10.1162/neco.1997.9.2.279 | pp. 279-304

We show that networks of relatively realistic mathematical models for biological neurons in principle can simulate arbitrary feedforward sigmoidal neural nets in a way that has previously not been considered. This new approach is based on temporal coding by single spikes (respectively by the timing of synchronous firing in pools of neurons) rather than on the traditional interpretation of analog variables in terms of firing rates. The resulting new simulation is substantially faster and hence more consistent with experimental results about the maximal speed of information processing in cortical neural systems. As a consequence we can show that networks of noisy spiking neurons are "universal approximators" in the sense that they can approximate with regard to temporal coding any given continuous function of several variables. This result holds for a fairly large class of schemes for coding analog variables by firing times of spiking neurons. This new proposal for the possible organization of computations in networks of spiking neurons systems has some interesting consequences for the type of learning rules that would be needed to explain the self-organization of such networks. Finally, the fast and noise-robust implementation of sigmoidal neural nets by temporal coding points to possible new ways of implementing feedforward and recurrent sigmoidal neural nets with pulse stream VLSI.

25. Computing with the Leaky Integrate-and-Fire Neuron: Logarithmic Computation and Multiplication

Doron Tal, Eric L. Schwartz

https://doi.org/10.1162/neco.1997.9.2.305 | pp. 305-318

The leaky integrate-and-fire (LIF) model of neuronal spiking (Stein 1967) provides an analytically tractable formalism of neuronal firing rate in terms of a neuron's membrane time constant, threshold, and refractory period. LIF neurons have mainly been used to model physiologically realistic spike trains, but little application of the LIF model appears to have been made in explicitly computational contexts. In this article, we show that the transfer function of a LIF neuron provides, over a wide-parameter range, a compressive nonlinearity sufficiently close to that of the logarithm so that LIF neurons can be used to multiply neural signals by mere addition of their outputs yielding the logarithm of the product. A simulation of the LIF multiplier shows that under a wide choice of parameters, a LIF neuron can log-multiply its inputs to within a 5% relative error.

26. Effect of Delay on the Boundary of the Basin of Attraction in a Self-Excited Single Graded-Response Neuron

Khashayar Pakdaman, C. P. Malta, C. Grotta-Ragazzo, J.-F. Vibert

https://doi.org/10.1162/neco.1997.9.2.319 | pp. 319-336

Little attention has been paid in the past to the effects of interunit transmission delays (representing axonal and synaptic delays) on the boundary of the basin of attraction of stable equilibrium points in neural networks. As a first step toward a better understanding of the influence of delay, we study the dynamics of a single graded-response neuron with a delayed excitatory self-connection. The behavior of this system is representative of that of a family of networks composed of graded-response neurons in which most trajectories converge to stable equilibrium points for any delay value. It is shown that changing the delay modifies the "location" of the boundary of the basin of attraction of the stable equilibrium points without affecting the stability of the equilibria. The dynamics of trajectories on the boundary are also delay dependent and influence the transient regime of trajectories within the adjacent basins. Our results suggest that when dealing with networks with delay, it is important to study not only the effect of the delay on the asymptotic convergence of the system but also on the boundary of the basins of attraction of the equilibria.

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27. Shattering All Sets of ‘k’ Points in “General Position” Requires (k — 1)/2 Parameters

Eduardo D. Sontag

https://doi.org/10.1162/neco.1997.9.2.337 | pp. 337-348

For classes of concepts defined by certain classes of analytic functions depending on n parameters, there are nonempty open sets of samples of length 2n + 2 that cannot be shattered. A slighly weaker result is also proved for piecewise-analytic functions. The special case of neural networks is discussed.

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28. Statistical Inference, Occam's Razor, and Statistical Mechanics on the Space of Probability Distributions

Vijay Balasubramanian

https://doi.org/10.1162/neco.1997.9.2.349 | pp. 349-368

The task of parametric model selection is cast in terms of a statistical mechanics on the space of probability distributions. Using the techniques of low-temperature expansions, I arrive at a systematic series for the Bayesian posterior probability of a model family that significantly extends known results in the literature. In particular, I arrive at a precise understanding of how Occam's razor, the principle that simpler models should be preferred until the data justify more complex models, is automatically embodied by probability theory. These results require a measure on the space of model parameters and I derive and discuss an interpretation of Jeffreys' prior distribution as a uniform prior over the distributions indexed by a family. Finally, I derive a theoretical index of the complexity of a parametric family relative to some true distribution that I call the razor of the model. The form of the razor immediately suggests several interesting questions in the theory of learning that can be studied using the techniques of statistical mechanics.

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29. Bias/Variance Analyses of Mixtures-of-Experts Architectures

Robert A. Jacobs

https://doi.org/10.1162/neco.1997.9.2.369 | pp. 369-383

This article investigates the bias and variance of mixtures-of-experts (ME) architectures. The variance of an ME architecture can be expressed as the sum of two terms: the first term is related to the variances of the expert networks that comprise the architecture and the second term is related to the expert networks' covariances. One goal of this article is to study and quantify a number of properties of ME architectures via the metrics of bias and variance. A second goal is to clarify the relationships between this class of systems and other systems that have recently been proposed. It is shown that in contrast to systems that produce unbiased experts whose estimation errors are uncorrelated, ME architectures produce biased experts whose estimates are negatively correlated.

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30. The Behavior of Forgetting Learning in Bidirectional Associative Memory

Chi-Sing Leung, Lai Wan Chan

https://doi.org/10.1162/neco.1997.9.2.385 | pp. 385-401

Forgetting learning is an incremental learning rule in associative memories. With it, the recent learning items can be encoded, and the old learning items will be forgotten. In this article, we analyze the storage behavior of bidirectional associative memory (BAM) under the forgetting learning. That is, “Can the most recent k learning item be stored as a fixed point?” Also, we discuss how to choose the forgetting constant in the forgetting learning such that the BAM can correctly store as many as possible of the most recent learning items. Simulation is provided to verify the theoretical analysis.

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31. Adaptive Encoding Strongly Improves Function Approximation with CMAC

Martin Eldracher, Alexander Staller, R. Pompl

https://doi.org/10.1162/neco.1997.9.2.403 | pp. 403-417

The Cerebellar Model Arithmetic Computer (CMAC) (Albus 1981) is well known as a good function approximator with local generalization abilities. Depending on the smoothness of the function to be approximated, the resolution as the smallest distinguishable part of the input domain plays a crucial role. If the binary quantizing functions in CMAC are dropped in favor of more general, continuous-valued functions, much better results in function approximation for smooth functions are obtained in shorter training time with less memory consumption. For functions with discontinuities, we obtain a further improvement by adapting the continuous encoding proposed in Eldracher and Geiger (1994) for difficult-to-approximate areas. Based on the already far better function approximation capability on continuous functions with a fixed topologically distributed encoding scheme in CMAC (Eldracher et al. 1994), we present the better results in learning a two-valued function with discontinuity using this adaptive topologically distributed encoding scheme in CMAC.

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32. An Analog Memory Circuit for Spiking Silicon Neurons

John G. Elias, D.P.M. Northmore, Wayne C. Westerman

https://doi.org/10.1162/neco.1997.9.2.419 | pp. 419-440

A simple circuit is described that functions as an analog memory whose state and dynamics are directly controlled by pulsatile inputs. The circuit has been incorporated into a silicon neuron with a spatially extensive dendritic tree as a means of controlling the spike firing threshold of an integrate-and-fire soma. Spiking activity generated by the neuron itself and by other units in a network can thereby regulate the neuron's excitability over time periods ranging from milliseconds to many minutes. Experimental results are presented showing applications to temporal edge sharpening, bistable behavior, and a network that learns in the manner of classical conditioning.

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33. Average-Case Learning Curves for Radial Basis Function Networks

Sean B. Holden, Mahesan Niranjan

https://doi.org/10.1162/neco.1997.9.2.441 | pp. 441-460

The application of statistical physics to the study of the learning curves of feedforward connectionist networks has to date been concerned mostly with perceptron-like networks. Recent work has extended the theory to networks such as committee machines and parity machines, and an important direction for current and future research is the extension of this body of theory to further connectionist networks. In this article, we use this formalism to investigate the learning curves of gaussian radial basis function networks (RBFNs) having fixed basis functions. (These networks have also been called generalized linear regression models.) We address the problem of learning linear and nonlinear, realizable and unrealizable, target rules from noise-free training examples using a stochastic training algorithm. Expressions for the generalization error, defined as the expected error for a network with a given set of parameters, are derived for general gaussian RBFNs, for which all parameters, including centers and spread parameters, are adaptable. Specializing to the case of RBFNs with fixed basis functions (basis functions having parameters chosen without reference to the training examples), we then study the learning curves for these networks in the limit of high temperature.

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34. A Sequential Learning Scheme for Function Approximation Using Minimal Radial Basis Function Neural Networks

Lu Yingwei, N. Sundararajan, P. Saratchandran

https://doi.org/10.1162/neco.1997.9.2.461 | pp. 461-478

This article presents a sequential learning algorithm for function approximation and time-series prediction using a minimal radial basis function neural network (RBFNN). The algorithm combines the growth criterion of the resource-allocating network (RAN) of Platt (1991) with a pruning strategy based on the relative contribution of each hidden unit to the overall network output. The resulting network leads toward a minimal topology for the RBFNN. The performance of the algorithm is compared with RAN and the enhanced RAN algorithm of Kadirkamanathan and Niranjan (1993) for the following benchmark problems: (1) hearta from the benchmark problems database PROBEN1, (2) Hermite polynomial, and (3) Mackey-Glass chaotic time series. For these problems, the proposed algorithm is shown to realize RBFNNs with far fewer hidden neurons with better or same accuracy.

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35. A Neuro-Mimetic Dynamic Scheduling Algorithm for Control: Analysis and Applications

Harpreet S. Kwatra, Francis J. Doyle, Ilya A. Rybak, James S. Schwaber

https://doi.org/10.1162/neco.1997.9.3.479 | pp. 479-502

A simple neuronal network model of the baroreceptor reflex is analyzed. From a control perspective, the analysis suggests a dynamic scheduled control mechanisms by which the baroreflex may perform regulation of the blood pressure. The main objectives of this work are to investigate the static and dynamic response characteristics of the single neurons and the network, to analyze the neuromimetic dynamic scheduled control function of the model, and to apply the algorithm to nonlinear process control problems. The dynamic scheduling activity of the network is exploited in two control architectures. Control structure I is drawn directly from the present model of the baroreceptor reflex. An application of this structure for level control in a conical tank is described. Control structure II employs an explicit set point to determine the feedback error. The performance of this control structure is illustrated on a nonlinear continuous stirred tank reactor with van de Vusse kinetics. The two case studies validate the dynamic scheduled control approach for nonlinear process control applications.

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36. Conductance-Based Integrate-and-Fire Models

Alain Destexhe

https://doi.org/10.1162/neco.1997.9.3.503 | pp. 503-514

A conductance-based model of Na+ and K+ currents underlying action potential generation is introduced by simplifying the quantitative model of Hodgkin and Huxley (HH). If the time course of rate constants can be approximated by a pulse, HH equations can be solved analytically. Pulse-based (PB) models generate action potentials very similar to the HH model but are computationally faster. Unlike the classical integrate-and-fire (IAF) approach, they take into account the changes of conductances during and after the spike, which have a determinant influence in shaping neuronal responses. Similarities and differences among PB, IAF, and HH models are illustrated for three cases: high-frequency repetitive firing, spike timing following random synaptic inputs, and network behavior in the presence of intrinsic currents.

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37. A Simple Model of Transmitter Release and Facilitation

Richard Bertram

https://doi.org/10.1162/neco.1997.9.3.515 | pp. 515-523

We describe a model of synaptic transmitter release and presynaptic facilitation that is based on activation of release sites by single Ca2+ microdomains. Facilitation is due to Ca2+ that remains bound to release sites between impulses. This model is inherently stochastic, but deterministic equations can be derived for the mean release. The number of equations required to describe the mean release is small, so it is practical to use the model with models of neuronal electrical activity to investigate the effects of different input spike patterns on presynaptic facilitation. We use it in conjunction with a model of dopamine-secreting neurons of the basal ganglia to demonstrate that transmitter release is greater when the neuron bursts than when it spikes continuously, due to the greater facilitation generated by the bursting impulse pattern. Finally, a minimal form of the model is described that is coupled to simple models of postsynaptic receptors and

passive membrane to compute the postsynaptic voltage response to a train of presynaptic stimuli. This form of the model is appropriate for neural network simulations.

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38. Functional Periodic Intracortical Couplings Induced by Structured Lateral Inhibition in a Linear Cortical Network

Silvio P. Sabatini, G. Bisio, Luigi Raffo

https://doi.org/10.1162/neco.1997.9.3.525 | pp. 525-531

The spatial organization of cortical axon and dendritic fields could be an interesting structural paradigm to obtain a functional specificity without postulating highly specific feedforward connections. In this article, we investigate the functional implications of recurrent intracortical inhibition when it occurs through clustered medium-range interconnection schemes (Wörgötter & Koch, 1991; Somogyi, 1989; Kritzer, Cowey, & Somogyi, 1992). Moreover, the interaction between the inhibitory schemes and visual orientation maps is explored. Assuming linearity, we show that clustered inhibitory mechanisms can trigger a propagation process that allows the development of extra (i.e., induced) interactions among the cortical sites involved in the recurrent loops. In addition, we point out how these interactions functionally modify the response of cortical simple cells and yield to highly structured Gabor-like receptive fields. This study should be considered not as a realistic biological model of the primary visual cortex but as an attempt to explain possible computational principles related to intracortical connectivity and to the underlying single-cell properties.

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39. Possible Roles of Spontaneous Waves and Dendritic Growth for Retinal Receptive Field Development

Pierre■Yves Burgi, Norberto M. Grzywacz

https://doi.org/10.1162/neco.1997.9.3.533 | pp. 533-553

Several models of cortical development postulate that a Hebbian process fed by spontaneous activity amplifies orientation biases occurring randomly in early wiring, to form orientation selectivity. These models are not applicable to the development of retinal orientation selectivity, since they neglect the polarization of the retina's poorly branched early dendritic trees and the wavelike organization of the retina's early noise. There is now evidence that dendritic polarization and spontaneous waves are key in the development of retinal receptive fields. When models of cortical development are modified to take these factors into account, one obtains a model of retinal development in which early dendritic polarization is the seed of orientation selectivity, while the spatial extent of spontaneous waves controls the spatial profile of receptive fields and their tendency to be isotropic.

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40. Singularities in Primate Orientation Maps

Klaus Obermayer, Gary G. Blasdel

https://doi.org/10.1162/neco.1997.9.3.555 | pp. 555-575

We report the results of an analysis of orientation maps in primate striate cortex with focus on singularities and their distribution. Data were obtained from squirrel monkeys and macaque monkeys of different ages.

We find the approximately 80% of singularities that are nearest neighbors have the opposite sign and that the spatial distribution of singularities differs significantly from a random distribution of points. We do not find evidence for consistent geometric patterns that singularities may form across the cortex. Except for a different overall alignment of orientation bands and different periods of repetition, maps obtained from different animals and different ages are found similar with respect to the measures used. Orientation maps are then compared with two different pattern models that are currently discussed in the literature: bandpass-filtered white noise, which accounts very well for the overall map structure, and the field analogy model, which specifies the orientation map by the location of singularities and their properties. The bandpass-filtered noise approach to orientation patterns correctly predicts the sign correlations between singularities and accounts for the deviations in the spatial distribution of singularities away from a random dot pattern. The field analogy model can account for the structure of certain local patches of the orientation map but not for the whole map. Neither of the models is completely satisfactory, and the structure of the orientation map remains to be fully explained.

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41. Topographic Receptive Fields and Patterned Lateral Interaction in a Self-Organizing Model of the Primary Visual Cortex

Joseph Sirosh, Risto Miikkulainen

https://doi.org/10.1162/neco.1997.9.3.577 | pp. 577-594

This article presents a self-organizing neural network model for the simultaneous and cooperative development of topographic receptive fields and lateral interactions in cortical maps. Both afferent and lateral connections adapt by the same Hebbian mechanism in a purely local and unsupervised learning process. Afferent input weights of each neuron self-organize into hill-shaped profiles, receptive fields organize topographically across the network, and unique lateral interaction profiles develop for each neuron. The model demonstrates how patterned lateral connections developed based on correlated activity and explains why lateral connection patterns closely follow receptive field properties such as ocular dominance.

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42. The Formation of Topographic Maps That Maximize the Average Mutual Information of the Output Responses to Noiseless Input Signals

Marc M. Van Hulle

https://doi.org/10.1162/neco.1997.9.3.595 | pp. 595-606

This article introduces an extremely simple and local learning rule for to pographic map formation. The rule, called the maximum entropy learning rule (MER), maximizes the unconditional entropy of the map's output for any type of input distribution. The aim of this article is to show that MER is a viable strategy for building topographic maps that maximize the average mutual information of the output responses to noiseless input signals when only input noise and noise-added input signals are available.

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43. Self-Organization of Firing Activities in Monkey's Motor Cortex: Trajectory Computation from Spike Signals

Siming Lin, Jennie Si, Andrew B. Schwartz

https://doi.org/10.1162/neco.1997.9.3.607 | pp. 607-621

The population vector method has been developed to combine the simultaneous direction-related activities of a population of motor cortical neurons to predict the trajectory of the arm movement. In this article, we consider a self-organizing model of a neural representation of the arm trajectory based on neuronal discharge rates. As self-organizing feature map (SOFM) is used to select the optimal set of weights in the model to determine the contribution of an individual neuron to an overall movement representation. The correspondence between movement directions and discharge patterns of the motor cortical neurons is established in the output map. The topology-preserving property of the SOFM is used to analyze the recorded data of a behaving monkey. The data used in this analysis were taken while the monkey was tracing spirals and doing center-->out movements. The arm trajectory could be well predicted using such a statistical model based on the motor cortex neuronal firing information. The SOFM method is compared with the population vector method, which extracts information related to trajectory by assuming that each cell has a fixed preferred direction during the task. This implies that these cells are acting along lines labeled only for direction. However, extradirectional information is carried in these cell responses. The SOFM has the capability of extracting not only direction-related information but also other parameters that are consistently represented in the activity of the recorded population of cells.

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44. Hyperparameter Selection for Self-Organizing Maps

Akio Utsugi

https://doi.org/10.1162/neco.1997.9.3.623 | pp. 623-635

The self-organizing map (SOM) algorithm for finite data is derived as an approximate maximum a posteriori estimation algorithm for a gaussian mixture model with a gaussian smoothing prior, which is equivalent to a generalized deformable model (GDM). For this model, objective criteria for selecting hyperparameters are obtained on the basis of empirical Bayesian estimation and cross-validation, which are representative model selection methods. The properties of these criteria are compared by simulation experiments. These experiments show that the cross-validation methods favor more complex structures than the expected log likelihood supports, which is a measure of compatibility between a model and data distribution. On the other hand, the empirical Bayesian methods have the opposite bias.

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45. Supervised Networks That Self-Organize Class Outputs

Ramesh R. Sarukkai

https://doi.org/10.1162/neco.1997.9.3.637 | pp. 637-648

Supervised, neural network, learning algorithms have proved very successful at solving a variety of learning problems; however, they suffer from a common problem of requiring explicit output labels. In this article, it is shown that pattern classification can be achieved, in a multilayered, feedforward, neural network, without requiring explicit output labels, by a process of supervised self-organization. The class projection is achieved by optimizing appropriate within-class uniformity and between-class discernibility criteria. The mapping function and the class labels are developed together iteratively using the derived self organizing backpropagation algorithm. The ability of the self-organizing network to generalize on unseen data is also experimentally evaluated on real data sets and compares favorably with the traditional labeled supervision with neural networks. In addition, interesting features emerge out of the proposed self-organizing

supervision, which are absent in conventional approaches.

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46. How Well Can We Estimate the Information Carried in Neuronal Responses from Limited Samples?

David Golomb, John Hertz, Stefano Panzeri, Alessandro Treves, Barry J. Richmond

https://doi.org/10.1162/neco.1997.9.3.649 | pp. 649-665

It is difficult to extract the information carried by neuronal responses about a set of stimuli because limited data samples result in biased estimates. Recently two improved procedures have been developed to calculate information from experimental results: a binning-and-correcting procedure and a neural network procedure. We have used data produced from a model of the spatiotemporal receptive fields of parvocellular and magnocellular lateral geniculate neurons to study the performance of these methods as a function of the number of trials used. Both procedures yield accurate results for one-dimensional neuronal codes. They can also be used to produce a reasonable estimate of the extra information in a three-dimensional code, in this instance, within 0.05-0.1 bit of the asymptotically calculated value--about 10% of the total transmitted information. We believe that this performance is much more accurate than previous procedures.

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47. Covariance Learning of Correlated Patterns in Competitive Networks

Ali A. Minai

https://doi.org/10.1162/neco.1997.9.3.667 | pp. 667-681

Covariance learning is a powerful type of Hebbian learning, allowing both potentiation and depression of synaptic strength. It is used for associative memory in feedforward and recurrent neural network paradigms. This article describes a variant of covariance learning that works particularly well for correlated stimuli in feedforward networks with competitive K-of-N firing. The rule, which is nonlinear, has an intuitive mathematical interpretation, and simulations presented in this article demonstrate its utility.

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48. A Mobile Robot That Learns Its Place

Sageev Oore, Geoffrey E. Hinton, Gregory Dudek

https://doi.org/10.1162/neco.1997.9.3.683 | pp. 683-699

We show how a neural network can be used to allow a mobile robot to derive an accurate estimate of its location from noisy sonar sensors and noisy motion information. The robot's model of its location is in the form of a probability distribution across a grid of possible locations. This distribution is updated using both the motion information and the predictions of a neural network that maps locations into likelihood distributions across possible sonar readings. By predicting sonar readings from locations, rather than vice versa, the robot can handle the very nongaussian noise in the sonar sensors. By using the constraint provided by the noisy motion information, the robot can use previous readings to improve its estimate of its current location. By treating the resulting estimates as if they were correct, the robot can learn the relationship between location and sonar readings without requiring an external supervision signal that specifies the actual location of the robot. It can learn to locate itself in a new environment with almost no supervision, and it can maintain its location ability even when the environment is nonstationary.

49. Similarity, Connectionism, and the Problem of Representation in Vision

Shimon Edelman, Sharon Duvdevani-Bar

https://doi.org/10.1162/neco.1997.9.4.701 | pp. 701-720

A representational scheme under which the ranking between represented similarities is isomorphic to the ranking between the corresponding shape similarities can support perfectly correct shape classification because it preserves the clustering of shapes according to the natural kinds prevailing in the external world. This article discusses the computational requirements of representation that preserves similarity ranks and points out the relative straightforwardness of its connectionist implementation.

50. Dynamic Model of Visual Recognition Predicts Neural Response Properties in the Visual Cortex

Rajesh P. N. Rao, Dana H. Ballard

https://doi.org/10.1162/neco.1997.9.4.721 | pp. 721-763

The responses of visual cortical neurons during fixation tasks can be significantly modulated by stimuli from beyond the classical receptive field. Modulatory effects in neural responses have also been recently reported in a task where a monkey freely views a natural scene. In this article, we describe a hierarchical network model of visual recognition that explains these experimental observations by using a form of the extended Kalman filter as given by the minimum description length (MDL) principle. The model dynamically combines input-driven bottom-up signals with expectation-driven top-down signals to predict current recognition state. Synaptic weights in the model are adapted in a Hebbian manner according to a learning rule also derived from the MDL principle. The resulting prediction-learning scheme can be viewed as implementing a form of expectation-maximization (EM) algorithm. The architecture of the model posits an active computational role of the reciprocal connections between adjoining visual cortical areas in determining neural response properties. In particular, the model demonstrates the possible role of feedback from higher cortical areas in mediating neurophysiological effects due to stimuli from beyond the classical receptive field. Simulations of the model are provided that help explain the experimental observations regarding neural responses in both free viewing and fixation conditions.

51. Correction to “Lower Bounds on VC-Dimension of Smoothly Parameterized Function Classes”

Wee Sun Lee, Peter L. Bartlett, Robert C. Williamson

https://doi.org/10.1162/neco.1997.9.4.765 | pp. 765-769

The earlier article gives lower bounds on the VC-dimension of various smoothly parameterized function classes. The results were proved by showing a relationship between the uniqueness of decision boundaries and the VC-dimension of smoothly parameterized function classes. The proof is incorrect; there is no such relationship under the conditions stated in the article. For the case of neural networks with tanh activation functions, we give an alternative proof of a lower bound for the VC-dimension proportional to the number of parameters, which holds even when the magnitude of the parameters is restricted to be arbitrarily small.

52. Lower Bound on VC-Dimension by Local Shattering

Yossi Erlich, Dan Chazan, Scott Petrack, Avraham Levy

https://doi.org/10.1162/neco.1997.9.4.771 | pp. 771-776

We show that the VC-dimension of a smoothly parameterized function class is not less than the dimension of any manifold in the parameter space, as long as distinct parameter values induce distinct decision boundaries. A similar theorem was published recently and used to introduce lower bounds on VC-dimension for several cases (Lee, Bartlett, & Williamson, 1995). This theorem is not correct, but our theorem could replace it for those cases and many other practical ones.

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53. SEEMORE: Combining Color, Shape, and Texture Histogramming in a Neurally Inspired Approach to Visual Object Recognition

Bartlett W. Mel

https://doi.org/10.1162/neco.1997.9.4.777 | pp. 777-804

Severe architectural and timing constraints within the primate visual system support the conjecture that the early phase of object recognition in the brain is based on a feedforward feature-extraction hierarchy. To assess the plausibility of this conjecture in an engineering context, a difficult three-dimensional object recognition domain was developed to challenge a pure feedforward, receptive-field-based recognition model called SEEMORE. SEEMORE is based on 102 viewpoint-invariant nonlinear filters that as a group are sensitive to contour, texture, and color cues. The visual domains consists of 100 real objects of many different types, including rigid (shovel), nonrigid (telephone cord), and statistical (maple leaf cluster) objects and photographs of complex scenes. Objects were individually presented in color video images under normal room lighting conditions. Based on 12 to 36 training views, SEEMORE was required to recognize unnormalized test views of objects that could vary in position, orientation in the image plane and in depth, and scale (factor of 2); for nonrigid objects, recognition was also tested under gross shape deformations. Correct classification performance on a test set consisting of 600 novel object views was 97 percent (chance was 1 percent) and was comparable for the subset of 15 nonrigid objects. Performance was also measured under a variety of image degradation conditions, including partial occlusion, limited clutter, color shift, and additive noise. Generalization behavior and classification errors illustrated the emergence of several striking natural shape categories that are not explicitly encoded in the dimensions of the feature space. It is concluded that in the light of the vast hardware resources available in the ventral stream of the primate visual system relative to those exercised here, the appealingly simple feature-space conjecture remains worthy of serious consideration as a neurobiological model.

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54. Image Segmentation Based on Oscillatory Correlation

DeLiang Wang, David Terman

https://doi.org/10.1162/neco.1997.9.4.805 | pp. 805-836

We study the image segmentation on the basis of locally excitatory, globally inhibitory oscillator networks (LEGION), whereby the phases of oscillators encode the binding of pixels. We introduce a lateral potential for each oscillators so that only oscillators with strong connections from their neighborhood can develop high potentials. Based on the concept of the lateral potential, a solution to remove noisy regions in an image is

proposed for LEGION, so that it suppresses the oscillators corresponding to noisy regions but without affecting those corresponding to major regions. We show that the resulting oscillator network separates an image into several major regions, plus a background consisting of all noisy regions, and we illustrate network properties by computer stimulation. The network exhibits a natural capacity in segmenting images. The oscillatory dynamics leads to a computer algorithm, which is applied successfully to segmenting real gray-level images. A number of issues regarding biological plausibility and perceptual organization are discussed. We argue that LEGION provides a novel and effective framework for image segmentation and figure-ground segregation.

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55. Stochastic Completion Fields: A Neural Model of Illusory Contour Shape and Salience

Lance R. Williams, David W. Jacobs

https://doi.org/10.1162/neco.1997.9.4.837 | pp. 837-858

We describe an algorithm- and representation-level theory of illusory contour shape and salience. Unlike previous theories, our model is derived from a single assumption: that the prior probability distribution of boundary completion shape can be modeled by a random walk in a lattice whose points are positions and orientations in the image plane (i.e., the space that one can reasonably assume is represented by neurons of the mammalian visual cortex). Our model does not employ numerical relaxation or other explicit minimization, but instead relies on the fact that the probability that a particle following a random walk will pass through a given position and orientation on a path joining two boundary fragments can be computed directly as the product of two vector-field convolutions. We show that for the random walk we define, the maximum likelihood paths are curves of least energy, that is, on average, random walks follow paths commonly assumed to model the shape of illusory contours. A computer model is demonstrated on numerous illusory contour stimuli from the literature.

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56. Local Parallel Computation of Stochastic Completion Fields

Lance R. Williams, David W. Jacobs

https://doi.org/10.1162/neco.1997.9.4.859 | pp. 859-881

We describe a local parallel method for computing the stochastic completion field introduced in the previous article (Williams and Jacobs, 1997). The stochastic completion field represents the likelihood that a completion joining two contour fragments passes through any given position and orientation in the image plane. It is based on the assumption that the prior probability distribution of completion shape can be modeled as a random walk in a lattice of discrete positions and orientations. The local parallel method can be interpreted as a stable finite difference scheme for solving the underlying Fokker-Planck equation identified by Mumford (1994). The resulting algorithm is significantly faster than the previously employed method, which relied on convolution with large-kernel filters computed by Monte Carlo simulation. The complexity of the new method is O (n 3 m), while that of the previous algorithm was O(n 4 m 2 (for an n × n image with m discrete orientations). Perhaps most significant, the use of a local method allows us to model the probability distribution of completion shape using stochastic processes that are neither homogeneous nor isotropic. For example, it is possible to modulate particle decay rate by a directional function of local image brightnesses (i.e., anisotropic decay). The effect is that illusory contours can be made to respect the

local image brightness structure. Finally, we note that the new method is more plausible as a neural model since (1) unlike the previous method, it can be computed in a sparse, locally connected network, and (2) the network dynamics are consistent with psychophysical measurements of the time course of illusory contour formation.

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57. Optimal, Unsupervised Learning in Invariant Object Recognition

Guy Wallis, Roland Baddeley

https://doi.org/10.1162/neco.1997.9.4.883 | pp. 883-894

A means for establishing transformation-invariant representations of objects is proposed and analyzed, in which different views are associated on the basis of the temporal order of the presentation of these views, as well as their spatial similarity. Assuming knowledge of the distribution of presentation times, an optimal linear learning rule is derived. Simulations of a competitive network trained on a character recognition task are then used t highlight the success of this learning rule in relation to simple Hebbian learning and to show that the theory can give accurate quantitative predictions for the optimal parameters for such networks.

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58. Activation Functions, Computational Goals, and Learning Rules for Local Processors with Contextual Guidance

Jim Kay, William A. Phillips

https://doi.org/10.1162/neco.1997.9.4.895 | pp. 895-910

Information about context can enable local processors to discover latent variables that are relevant to the context within which they occur, and it can also guide short-term processing. For example, Becker and Hinton (1992) have shown how context can guide learning, and Hummel and Biederman (1992) have shown how it can guide processing in a large neural net for object recognition. This article studies the basic capabilities of a local processor with two distinct classes of inputs: receptive field inputs that provide the primary drive and contextual inputs that modulate their effects. The contextual predictions are used to guide processing without confusing them with receptive field inputs. The processor's transfer function must therefore distinguish these two roles. Given these two classes of input, the information in the output can be decomposed into four disjoint components to provide a space of possible goals in which the unsupervised learning of Linsker (1988) and the internally supervised learning of Becker and Hinton (1992) are special cases. Learning rules are derived from an information-theoretic objective function, and simulations show that a local processor trained with these rules and using an appropriate activation function has the elementary properties required.

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59. Marr's Theory of the Neocortex as a Self-Organizing Neural Network

David Willshaw, John Hallam, Sarah Gingell, Soo Leng Lau

https://doi.org/10.1162/neco.1997.9.4.911 | pp. 911-936

Marr's proposal for the functioning of the neocortex (Marr, 1970) is the least known of his various theories for specific neural circuitries. He suggested that the neocortex learns by self-organization to extract the structure from the patterns of activity incident upon it. He proposed a feedforward neural network in which

the connections to the output cells (identified with the pyramidal cells of the neocortex) are modified by a mechanism of competitive learning. It was intended that each output cell comes to be selective for the input patterns from a different class and is able to respond to new patterns from the same class that have not been seen before. The learning rule that Marr proposed was underspecified, but a logical extension of the basic idea results in a synaptic learning rule in which the total amount of synaptic strength of the connections from each input ("presynaptic") cell is kept at a constant level. In contrast, conventional competitive learning involves rules of the "postsynaptic" type. The network learns by exploiting the structure that Marr assumed to exist within the ensemble of input patterns. For this case, analysis is possible that extends that carried out by Marr, which was restricted to the binary classification task. This analysis is presented here, together with results from computer simulations of different types of competitive learning mechanisms. The presynaptic mechanism is best known in the computational neuroscience literature. In neural network applications, it may be a more suitable mechanism of competitive learning than those normally considered.

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60. Hybrid Learning of Mapping and its Jacobian in Multilayer Neural Networks

Jeongwoo Lee, Jun-Ho Oh

https://doi.org/10.1162/neco.1997.9.5.937 | pp. 937-958

There are some learning problems for which a priori information, such as the Jacobian of mapping, is available in addition to input-output examples. This kind of information can be beneficial in neural network learning if it can be embedded into the network. This article is concerned with the method for learning the mapping and available Jacobian simultaneously. The basic idea is to minimize the cost function, which is composed of the mapping error and the Jacobian error. Prior to developing the Jacobian learning rule, we develop an explicit and general method for computing the Jacobian of the neural network using the parameters computed in error backpropagation. Then the Jacobian learning rule is derived. The method is simpler and more general than tangent propagation (Simard, Victorri, Le Cun, & Denker, 1992). Through hybridization of the error backpropagation and the Jacobian learning, the hybrid learning algorithm is presented. The method shows good performance in accelerating the learning speed and improving generalization. And through computer experiments, it is shown that using the Jacobian synthesized from noise-corrupted data can accelerate learning speed.

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61. The Joint Development of Orientation and Ocular Dominance: Role of Constraints

Christian Piepenbrock, Helge Ritter, Klaus Obermayer

https://doi.org/10.1162/neco.1997.9.5.959 | pp. 959-970

Correlation-based learning (CBL) has been suggested as the mechanism that underlies the development of simple-cell receptive fields in the primary visual cortex of cats, including orientation preference (OR) and ocular dominance (OD) (Linsker, 1986; Miller, Keller, & Stryker, 1989). CBL has been applied successfully to the development of OR and OD individually (Miller, Keller, & Stryker, 1989; Miller, 1994; Miyashita & Tanaka, 1991; Erwin, Obermayer, & Schulten, 1995), but the conditions for their joint development have not been studied (but see Erwin & Miller, 1995, for independent work on the same question) in contrast to

competitive Hebbian models (Obermayer, Blasdel, & Schulten, 1992). In this article, we provide insight into why this has been the case: OR and OD decouple in symmetric CBL models, and a joint development of OR and OD is possible only in a parameter regime that depends on nonlinear mechanisms.

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62. Physiological Gain Leads to High ISI Variability in a Simple Model of a Cortical Regular Spiking Cell

Todd W. Troyer, Kenneth D. Miller

https://doi.org/10.1162/neco.1997.9.5.971 | pp. 971-983

To understand the interspike interval (ISI) variability displayed by visual cortical neurons (Softky & Koch, 1993), it is critical to examine the dynamics of their neuronal integration, as well as the variability in their synaptic input current. Most previous models have focused on the latter factor. We match a simple integrate-and-fire model to the experimentally measured integrative properties of cortical regular spiking cells (McCormick, Connors, Lighthall, & Prince, 1985). After setting RC parameters, the post-spike voltage reset is set to match experimental measurements of neuronal gain (obtained from in vitro plots of firing frequency versus injected current). Examination of the resulting model leads to an intuitive picture of neuronal integration that unifies the seemingly contradictory 1/square root of N and random walk pictures that have previously been proposed. When ISIs are dominated by postspike recovery, 1/square root of N arguments hold and spiking is regular; after the "memory" of the last spike becomes negligible, spike threshold crossing is caused by input variance around a steady state and spiking is Poisson. In integrate-and-fire neurons matched to cortical cell physiology, steady-state behavior is predominant, and ISIs are highly variable at all physiological firing rates and for a wide range of inhibitory and excitatory inputs.

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63. Role of Temporal Integration and Fluctuation Detection in the Highly Irregular Firing of a Leaky Integrator Neuron Model with Partial Reset

Guido Bugmann, Chris Christodoulou, John G. Taylor

https://doi.org/10.1162/neco.1997.9.5.985 | pp. 985-1000

Partial reset is a simple and powerful tool for controlling the irregularity of spike trains fired by a leaky integrator neuron model with random inputs. In particular, a single neuron model with a realistic membrane time constant of 10 ms can reproduce the highly irregular firing of cortical neurons reported by Softky and Koch (1993). In this article, the mechanisms by which partial reset affects the firing pattern are investigated. Itisshown theoretically that partial reset is equivalent to the use of a time-dependent threshold, similar to a technique proposed by Wilbur and Rinzel (1983) to produce high irregularity. This equivalent model allows establishing that temporal integration and fluctuation detection can coexist and cooperate to cause highly irregular firing. This study also reveals that reverse correlation curves cannot be used reliably to assess the causes of firing. For instance, they do not reveal temporal integration when it takes place. Further, the peak near time zero does not always indicate coincidence detection. An alternative qualitative method is proposed here for that later purpose. Finally, it is noted that as the reset becomes weaker, the firing pattern shows a progressive transition from regular firing, to random, to temporally clustered, and eventually to bursting firing. Concurrently the slope of the transfer function increases. Thus, simulations suggest a correlation between high gain and highly irregular firing.

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64. Shunting Inhibition Does Not Have a Divisive Effect on Firing Rates

Gary R. Holt, Christof Koch

https://doi.org/10.1162/neco.1997.9.5.1001 | pp. 1001-1013

Shunting inhibition, a conductance increase with a reversal potential close to the resting potential of the cell, has been shown to have a divisive effect on subthreshold excitatory postsynaptic potential amplitudes. It has therefore been assumed to have the same divisive effect on firing rates. We show that shunting inhibition actually has a subtractive effect on the firing rate in most circumstances. Averaged over several interspike intervals, the spiking mechanism effectively clamps the somatic membrane potential to a value significantly above the resting potential, so that the current through the shunting conductance is approximately independent of the firing rate. This leads to a subtractive rather than a divisive effect. In addition, at distal synapses, shunting inhibition will also have an approximately subtractive effect if the excitatory conductance is not small compared to the inhibitory conductance. Therefore regulating a cell's passive membrane conductance-for instance, via massive feedback-is not an adequate mechanism for normalizing or scaling its output.

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65. Reduction of the Hodgkin-Huxley Equations to a Single-Variable Threshold Model

Werner M. Kistler, Wulfram Gerstner, J. Leo van Hemmen

https://doi.org/10.1162/neco.1997.9.5.1015 | pp. 1015-1045

It is generally believed that a neuron is a threshold element that fires when some variable u reaches a threshold. Here we pursue the question of whether this picture can be justified and study the four-dimensional neuron model of Hodgkin and Huxley as a concrete example. The model is approximated by a response kernel expansion in terms of a single variable, the membrane voltage. The first-order term is linear in the input and its kernel has the typical form of an elementary postsynaptic potential. Higher-order kernels take care of nonlinear interactions between input spikes. In contrast to the standard Volterra expansion, the kernels depend on the firing time of the most recent output spike. In particular, a zero-order kernel that describes the shape of the spike and the typical after-potential is included. Our model neuron fires if the membrane voltage, given by the truncated response kernel expansion, crosses a threshold. The threshold model is tested on a spike train generated by the Hodgkin-Huxley model with a stochastic input current. We find that the threshold model predicts 90 percent of the spikes correctly. Our results show that, to good approximation, the description of a neuron as a threshold element can indeed be justified.

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66. Noise Adaptation in Integrate-and-Fire Neurons

Michael E. Rudd, Lawrence G. Brown

https://doi.org/10.1162/neco.1997.9.5.1047 | pp. 1047-1069

The statistical spiking response of an ensemble of identically prepared stochastic integrate-and-fire neurons to a rectangular input current plus gaussian white noise is analyzed. It is shown that, on average, integrate-and-fire neurons adapt to the root-mean-square noise level of their input. This phenomenon is referred to as noise adaptation. Noise adaptation is characterized by a decrease in the average neural firing rate and an accompanying decrease in the average value of the generator potential, both of which can be

attributed to noise-induced resets of the generator potential mediated by the integrate-and-fire mechanism. A quantitative theory of noise adaptation in stochastic integrate-and-fire neurons is developed. It is shown that integrate-and-fire neurons, on average, produce transient spiking activity whenever there is an increase in the level of their input noise. This transient noise response is either reduced or eliminated over time, depending on the parameters of the model neuron. Analytical methods are used to prove that nonleaky integrate-and-fire neurons totally adapt to any constant input noise level, in the sense that their asymptotic spiking rates are independent of the magnitude of their input noise. For leaky integrate-and-fire neurons, the long-run noise adaptation is not total, but the response to noise is partially eliminated. Expressions for the probability density function of the generator potential and the first two moments of the potential distribution are derived for the particular case of a nonleaky neuron driven by gaussian white noise of mean zero and constant variance. The functional significance of noise adaptation for the performance of networks comprising integrate-and-fire neurons is discussed.

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67. Paradigmatic Working Memory (Attractor) Cell in IT Cortex

Daniel J. Amit, Stefano Fusi, Volodya Yakovlev

https://doi.org/10.1162/neco.1997.9.5.1071 | pp. 1071-1092

We discuss paradigmatic properties of the activity of single cells comprising an attractor-a developed stable delay activity distribution. To demonstrate these properties and a methodology for measuring their values, we present a detailed account of the spike activity recorded from a single cell in the inferotemporal cortex of a monkey performing a delayed match-to-sample (DMS) task of visual images. In particular, we discuss and exemplify (1) the relation between spontaneous activity and activity immediately preceding the first stimulus in each trial during a series of DMS trials, (2) the effect on the visual response (i.e., activity during stimulation) of stimulus degradation (moving in the space of IT afferents), (3) the behavior of the delay activity (i.e., activity following visual stimulation) under stimulus degradation (attractor dynamics and the basin of attraction), and (4) the propagation of information between trials-the vehicle for the formation of (contextual) correlations by learning a fixed stimulus sequence (Miyashita, 1988). In the process of the discussion and demonstration, we expose effective tools for the identification and characterization of attractor dynamics.

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68. Noise Injection: Theoretical Prospects

Yves Grandvalet, Stéphane Canu, Stéphane Boucheron

https://doi.org/10.1162/neco.1997.9.5.1093 | pp. 1093-1108

Noise injection consists of adding noise to the inputs during neural network training. Experimental results suggest that it might improve the generalization ability of the resulting neural network. A justification of this improvement remains elusive: describing analytically the average perturbed cost function is difficult, and controlling the fluctuations of the random perturbed cost function is hard. Hence, recent papers suggest replacing the random perturbed cost by a (deterministic) Taylor approximation of the average perturbed cost function. This article takes a different stance: when the injected noise is gaussian, noise injection is naturally connected to the action of the heat kernel. This provides indications on the relevance domain of traditional Taylor expansions and shows the dependence of the quality of Taylor approximations on global smoothness properties of neural networks under consideration. The connection between noise injection and heat kernel

also enables controlling the fluctuations of the random perturbed cost function. Under the global smoothness assumption, tools from gaussian analysis provide bounds on the tail behavior of the perturbed cost. This finally suggests that mixing input perturbation with smoothness-based penalization might be profitable.

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69. The Faulty Behavior of Feedforward Neural Networks with Hard-Limiting Activation Function

Zhiyu Tian, Ting-Ting Y. Lin, Shiyuan Yang, Shibai Tong

https://doi.org/10.1162/neco.1997.9.5.1109 | pp. 1109-1126

With the progress in hardware implementation of artificial neural networks, the ability to analyze their faulty behavior has become increasingly important to their diagnosis, repair, reconfiguration, and reliable application. The behavior of feedforward neural networks with hard limiting activation function under stuck-at faults is studied in this article. It is shown that the stuck-at-M faults have a larger effect on the network's performance than the mixed stuck-at faults, which in turn have a larger effect than that of stuck-at-0 faults. Furthermore, the fault-tolerant ability of the network decreases with the increase of its size for the same percentage of faulty interconnections. The results of our analysis are validated by Monte-Carlo simulations.

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70. Analysis of Dynamical Recognizers

Alan Blair, Jordan Pollack

https://doi.org/10.1162/neco.1997.9.5.1127 | pp. 1127-1142

Pollack (1991) demonstrated that second-order recurrent neural networks can act as dynamical recognizers for formal languages when trained on positive and negative examples, and observed both phase transitions in learning and interacted function system-like fractal state sets. Follow on work focused mainly on the extraction and minimization of a finite state automaton (FSA) from the trained network. However, such networks are capable of inducing languages that are not regular and therefore not equivalent to any FSA. Indeed, it may be simpler for a small network to fit its training data by inducing such a nonregular language. But when is the network's language not regular? In this article, using a low-dimensional network capable of learning all the Tomita data sets, we present an empirical method for testing whether the language induced by the network is regular. We also provide a detailed "-machine analysis of trained networks for both regular and nonregular languages.

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71. A Bound on the Error of Cross Validation Using the Approximation and Estimation Rates, with Consequences for the Training-Test Split

Michael Kearns

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We give a theoretical and experimental analysis of the generalization error of cross validation using two natural measures of the problem under consideration. The approximation rate measures the accuracy to which the target function can be ideally approximated as a function of the number of parameters, and thus captures the complexity of the target function with respect to the hypothesis model. The estimation rate measures the deviation between the training and generalization errors as a function of the number of

parameters, and thus captures the extent to which the hypothesis model suffers from overfitting. Using these two measures, we give a rigorous and general bound on the error of the simplest form of cross validation. The bound clearly shows the dangers of making γ —the fraction of data saved for testing—too large or too small. By optimizing the bound with respect to γ, we then argue that the following qualitative properties of cross-validation behavior should be quite robust to significant changes in the underlying model selection problem: When the target function complexity is small compared to the sample size, the performance of cross validation is relatively insensitive to the choice of γ. The importance of choosing γ optimally increases, and the optimal value for γ decreases, as the target function becomes more complex relative to the sample size. There is nevertheless a single fixed value for γ that works nearly optimally for a wide range of target function complexity.

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72. A bound on the error of cross validation using the approximation and estimation rates, with

Michael Kearns

(no DOI) | pp. 1143-1161

No abstract available.

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73. Averaging Regularized Estimators

Michiaki Taniguchi, Volker Tresp

https://doi.org/10.1162/neco.1997.9.5.1163 | pp. 1163-1178

We compare the performance of averaged regularized estimators. We show that the improvement in performance that can be achieved by averaging depends critically on the degree of regularization which is used in training the individual estimators. We compare four different averaging approaches: simple averaging, bagging, variance-based weighting, and variance-based bagging. In any of the averaging methods, the greatest degree of improvement—if compared to the individual estimators—is achieved if no or only a small degree of regularization is used. Here, variance-based weighting and variance-based bagging are superior to simple averaging or bagging. Our experiments indicate that better performance for both individual estimators and for averaging is achieved in combination with regularization. With increasing degrees of regularization, the two bagging-based approaches (bagging and variance-based bagging) outperform the individual estimators, simple averaging, and variance-based weighting. Bagging and variance-based bagging seem to be the overall best combining methods over a wide range of degrees of regularization.

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74. The NEURON Simulation Environment

Michael L. Hines, Nicholas T. Carnevale

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The moment-to-moment processing of information by the nervous system involves the propagation and interaction of electrical and chemical signals that are distributed in space and time. Biologically realistic modeling is needed to test hypotheses about the mechanisms that govern these signals and how nervous

system function emerges from the operation of these mechanisms. The NEURON simulation program provides a powerful and flexible environment for implementing such models of individual neurons and small networks of neurons. It is particularly useful when membrane potential is nonuniform and membrane currents are complex. We present the basic ideas that would help informed users make the most efficient use of NEURON.

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75. On Bias Plus Variance

David H. Wolpert

https://doi.org/10.1162/neco.1997.9.6.1211 | pp. 1211-1243

This article presents several additive corrections to the conventional quadratic loss bias-plus-variance formula. One of these corrections is appropriate when both the target is not fixed (as in Bayesian analysis) and training sets are averaged over (as in the conventional bias plus variance formula). Another additive correction casts conventional fixed-trainingset Bayesian analysis directly in terms of bias plus variance. Another correction is appropriate for measuring full generalization error over a test set rather than (as with conventional bias plus variance) error at a single point. Yet another correction can help explain the recent counterintuitive bias-variance decomposition of Friedman for zero-one loss. After presenting these corrections, this article discusses some other loss function-specific aspects of supervised learning. In particular, there is a discussion of the fact that if the loss function is a metric (e.g., zero-one loss), then there is bound on the change in generalization error accompanying changing the algorithm's guess from h 1 to h 2 , a bound that depends only on h 1 and h 2 and not on the target. This article ends by presenting versions of the bias-plus-variance formula appropriate for logarithmic and quadratic scoring, and then all the additive corrections appropriate to those formulas. All the correction terms presented are a covariance, between the learning algorithm and the posterior distribution over targets. Accordingly, in the (very common) contexts in which those terms apply, there is not a “bias-variance trade-off” or a “bias-variance dilemma,” as one often hears. Rather there is a bias-variance-covariance trade-off.

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76. Note on Free Lunches and Cross-Validation

Cyril Goutte

https://doi.org/10.1162/neco.1997.9.6.1245 | pp. 1245-1249

The “no-free-lunch” theorems (Wolpert & Macready, 1995) have sparked heated debate in the computational learning community. A recent communication (Zhu & Rohwer, 1996) attempts to demonstrate the inefficiency of cross-validation on a simple problem. We elaborate on this result by considering a broader class of cross-validation. When used more strictly, cross-validation can yield the expected results on simple examples.

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77. Gamma Oscillation Model Predicts Intensity Coding by Phase Rather than Frequency

Roger D. Traub, Miles A. Whittington, John G. R. Jefferys

https://doi.org/10.1162/neco.1997.9.6.1251 | pp. 1251-1264

Gamma-frequency electroencephalogram oscillations may be important for cognitive processes such as feature binding. Gamma oscillations occur in hippocampus in vivo during the theta state, following physiological sharp waves, and after seizures, and they can be evoked in vitro by tetanic stimulation. In neocortex, gamma oscillations occur under conditions of sensory stimulation as well as during sleep. After tetanic or sensory stimulation, oscillations in regions separated by several millimeters or more occur at the same frequency, but with phase lags ranging from less than 1 ms to 10 ms, depending on the conditions of stimulation. We have constructed a distributed network model of pyramidal cells and interneurons, based on a variety of experiments, that accounts for near-zero phase lag synchrony of oscillations over long distances (with axon conduction delays totaling 16 ms or more). Here we show that this same model can also account for fixed positive phase lags between nearby cell groups coexisting with near-zero phase lags between separated cell groups, a phenomenon known to occur in visual cortex. The model achieves this because interneurons fire spike doublets and triplets that have average zero phase difference throughout the network; this provides a temporal framework on which pyramidal cell phase lags can be superimposed, the lag depending on how strongly the pyramidal cells are excited.

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78. Multiunit Normalized Cross Correlation Differs from the Average Single-Unit Normalized Correlation

Purvis Bedenbaugh, George L. Gerstein

https://doi.org/10.1162/neco.1997.9.6.1265 | pp. 1265-1275

As the technology for simultaneously recording from many brain locations becomes more available, more and more laboratories are measuring the cross-correlation between single-neuron spike trains, and between composite spike trains derived from several undiscriminated cells recorded on a single electrode (multiunit clusters). The relationship between single-unit correlations and multiunit cluster correlations has not yet been fully explored. We calculated the normalized cross-correlation (NCC) between single-unit spike trains and between small clusters of units recorded in the rat somatosensory cortex. The NCC between small clusters of units was larger than the NCC between single units. To understand this result, we investigated the scaling of the NCC with the number of units in a cluster. Multiunit cross-correlation can be a more sensitive detector of neuronal relationship than single-unit cross-correlation. However, changes in multiunit cross-correlation are difficult to interpret uniquely because they depend on the number of cells recorded on each electrode and because they can arise from changes in the correlation between cells recorded on a single electrode or from changes in the correlation between cells recorded on two electrodes.

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79. A Simple Common Contexts Explanation for the Development of Abstract Letter Identities

Thad A. Polk, Martha J. Farah

https://doi.org/10.1162/neco.1997.9.6.1277 | pp. 1277-1289

Abstract letter identities (ALIs) are an early representation in visual word recognition that are specific to written language. They do not reflect visual or phonological features, but rather encode the identities of letters independent of case, font, sound, and so forth. How could the visual system come to develop such a representation? We propose that because many letters look similar regardless of case, font, and other characteristics, these provide common contexts for visually dissimilar uppercase and lowercase forms of

other letters (e.g., e between k and y in key and E in the visually similar context K-Y). Assuming that the distribution of words' relative frequencies is comparable in upper- and lowercase (that just as key is more frequent than pew, KEY is more frequent than PEW), these common contexts will also be similarly distributed in the two cases. We show how this statistical regularity could lead Hebbian learning to produce ALIs in a competitive architecture. We present a self-organizing artificial neural network that illustrates this idea and produces ALIs when presented with the most frequent words from a beginning reading corpus, as well as with artificial input.

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80. A Unifying Objective Function for Topographic Mappings

Geoffrey J. Goodhill, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1997.9.6.1291 | pp. 1291-1303

Many different algorithms and objective functions for topographic mappings have been proposed. We show that several of these approaches can be seen as particular cases of a more general objective function. Consideration of a very simple mapping problem reveals large differences in the form of the map that each particular case favors. These differences have important consequences for the practical application of topographic mapping methods.

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81. Faithful Representation of Separable Distributions

Juan K. Lin, David G. Grier, Jack D. Cowan

https://doi.org/10.1162/neco.1997.9.6.1305 | pp. 1305-1320

A geometric approach to data representation incorporating information theoretic ideas is presented. The task of finding a faithful representation, where the input distribution is evenly partitioned into regions of equal mass, is addressed. For input consisting of mixtures of statistically independent sources, we treat independent component analysis (ICA) as a computational geometry problem. First, we consider the separation of sources with sharply peaked distribution functions, where the ICA problem becomes that of finding high-density directions in the input distribution. Second, we consider the more general problem for arbitrary input distributions, where ICA is transformed into the task of finding an aligned equipartition. By modifying the Kohonen self-organized feature maps, we arrive at neural networks with local interactions that optimize coding while simultaneously performing source separation. The local nature of our approach results in networks with nonlinear ICA capabilities.

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82. Self-Organized Formation of Various Invariant-Feature Filters in the Adaptive-Subspace SOM

Teuvo Kohonen, Samuel Kaski, Harri Lappalainen

https://doi.org/10.1162/neco.1997.9.6.1321 | pp. 1321-1344

The adaptive-subspace self-organizing map (ASSOM) is a modular neural network architecture, the modules of which learn to identify input patterns subject to some simple transformations. The learning process is unsupervised, competitive, and related to that of the traditional SOM (self-organizing map). Each neural module becomes adaptively specific to some restricted class of transformations, and modules close

to each other in the network become tuned to similar features in an orderly fashion. If different transformations exist in the input signals, different subsets of ASSOM units become tuned to these transformation classes.

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83. Self-Organizing Map with Dynamical Node Splitting: Application to Handwritten Digit Recognition

Sung■Bae Cho

https://doi.org/10.1162/neco.1997.9.6.1345 | pp. 1345-1355

This article presents a simple yet elegant pattern recognizer based on a dynamic node-splitting scheme for the self-organizing map that can adapt its structure as well as its weights. The scheme makes use of a structure adaptation capability to place the nodes of prototype vectors into the pattern space accurately so as to make the decision boundaries as close to the class boundaries as possible. In order to show the performance of the proposed scheme, experiments with the unconstrained handwritten digit database of Concordia University in Canada were conducted. The proposed method for an incremental formation of feature maps is 96.05 percent of the recognition rate. In view of the elegant simplicity of the approach, the reported performance is remarkable and can stand up to one of the best results reported in the literature with the same database.

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84. Information Controller to Maximize and Minimize Information

Ryotaro Kamimura

https://doi.org/10.1162/neco.1997.9.6.1357 | pp. 1357-1380

We propose an information controller to maximize and simultaneously minimize information in neural networks. The controller can be used to improve generalization performance and interpret explicitly internal representations. An apparently contradictory operation of the simultaneous maximization and minimization is possible by assuming two kinds of information: collective information and individual information, representing collective activities of hidden units and activities of individual input-hidden connections, respectively. By maximizing the collective information and minimizing the individual information, simple networks can be generated in terms of the number of connections and number of hidden units. This method was applied to the inference of the maximum onset principle and the sonority principle of artificial languages. In these problems, it was shown that the individual information could be sufficiently minimized and simultaneously the collective information was maximized. Experimental results confirmed improved generalization performance by the suppression of overtraining. In addition, internal representations created by information maximization and minimization could easily be interpreted.

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85. Controlling Hidden Layer Capacity Through Lateral Connections

Kwabena Agyepong, Ravi Kothari

https://doi.org/10.1162/neco.1997.9.6.1381 | pp. 1381-1402

We investigate the effects of including selected lateral interconnections in a feedforward neural network. In a network with one hidden layer consisting of m hidden neurons labeled 1,2… m, hidden neuron j is connected

fully to the inputs, the outputs, and hidden neuron j + 1. As a consequence of the lateral connections, each hidden neuron receives two error signals: one from the output layer and one through the lateral interconnection. We show that the use of these lateral interconnections among the hidden-layer neurons facilitates controlled assignment of role and specialization of the hidden-layer neurons. In particular, we show that as training progresses, hidden neurons become progressively specialized—starting from the fringes (i.e., lower and higher numbered hidden neurons, e.g., 1, 2, m — 1 m) and leaving the neurons in the center of the hidden layer (i.e., hidden-layer neurons numbered close to m/2) unspecialized or functionally identical. Consequently, the network behaves like network growing algorithms without the explicit need to add hidden units, and like soft weight sharing due to functionally identical neurons in the center of the hidden layer. Experimental results from one classification and one function approximation problems are presented to illustrate selective specialization of the hidden-layer neurons. In addition, the improved generalization that results from a decrease in the effective number of free parameters is illustrated through a simple function approximation example and with a real-world data set. Besides the reduction in the number of free parameters, the localization of weight sharing may also allow for a method that allows procedural determination for the number of hidden-layer neurons required for a given learning task.

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86. Mean-Field Theory for Batched TD(λ)

Fernando J. Pineda

https://doi.org/10.1162/neco.1997.9.7.1403 | pp. 1403-1419

A representation-independent mean-field dynamics is presented for batched TD(λ). The task is learning to predict the outcome of an indirectly observed absorbing Markov process. In the case of linear representations, the discrete-time deterministic iteration is an affine map whose fixed point can be expressed in closed form without the assumption of linearly independent observation vectors. Batched linear TD(λ) is proved to converge with probability 1 for all λ. Theory and simulation agree on a random walk example.

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87. Redundancy Reduction and Independent Component Analysis: Conditions on Cumulants and Adaptive Approaches

Jean■Pierre Nadal, Néstor Parga

https://doi.org/10.1162/neco.1997.9.7.1421 | pp. 1421-1456

In the context of both sensory coding and signal processing, building factorized codes has been shown to be an efficient strategy. In a wide variety of situations, the signal to be processed is a linear mixture of statistically independent sources. Building a factorized code is then equivalent to performing blind source separation. Thanks to the linear structure of the data, this can be done, in the language of signal processing, by finding an appropriate linear filter, or equivalently, in the language of neural modeling, by using a simple feedforward neural network. In this article, we discuss several aspects of the source separation problem. We give simple conditions on the network output that, if satisfied, guarantee that source separation has been obtained. Then we study adaptive approaches, in particular those based on redundancy reduction and maximization of mutual information. We show how the resulting updating rules are related to the BCM theory of synaptic plasticity. Eventually we briefly discuss extensions to the case of nonlinear mixtures. Through out this article, we take care to put into perspective our work with other studies on source separation and

redundancy reduction. In particular we review algebraic solutions, pointing out their simplicity but also their drawbacks.

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88. Adaptive Online Learning Algorithms for Blind Separation: Maximum Entropy and Minimum Mutual Information

Howard H. Yang, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.1997.9.7.1457 | pp. 1457-1482

There are two major approaches for blind separation: maximum entropy (ME) and minimum mutual information (MMI). Both can be implemented by the stochastic gradient descent method for obtaining the demixing matrix. The MI is the contrast function for blind separation; the entropy is not. To justify the ME, the relation between ME and MMI is first elucidated by calculating the first derivative of the entropy and proving that the mean subtraction is necessary in applying the ME and at the solution points determined by the MI, the ME will not update the demixing matrix in the directions of increasing the cross-talking. Second, the natural gradient instead of the ordinary gradient is introduced to obtain efficient algorithms, because the parameter space is a Riemannian space consisting of matrices. The mutual information is calculated by applying the Gram-Charlier expansion to approximate probability density functions of the outputs. Finally, we propose an efficient learning algorithm that incorporates with an adaptive method of estimating the unknown cumulants. It is shown by computer simulation that the convergence of the stochastic descent algorithms is improved by using the natural gradient and the adaptively estimated cumulants.

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89. A Fast Fixed-Point Algorithm for Independent Component Analysis

Aapo Hyvärinen, Erkki Oja

https://doi.org/10.1162/neco.1997.9.7.1483 | pp. 1483-1492

We introduce a novel fast algorithm for independent component analysis, which can be used for blind source separation and feature extraction. We show how a neural network learning rule can be transformed into a fixedpoint iteration, which provides an algorithm that is very simple, does not depend on any user-defined parameters, and is fast to converge to the most accurate solution allowed by the data. The algorithm finds, one at a time, all nongaussian independent components, regardless of their probability distributions. The computations can be performed in either batch mode or a semiadaptive manner. The convergence of the algorithm is rigorously proved, and the convergence speed is shown to be cubic. Some comparisons to gradient-based algorithms are made, showing that the new algorithm is usually 10 to 100 times faster, sometimes giving the solution in just a few iterations.

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90. Dimension Reduction by Local Principal Component Analysis

Nandakishore Kambhatla, Todd K. Leen

https://doi.org/10.1162/neco.1997.9.7.1493 | pp. 1493-1516

Reducing or eliminating statistical redundancy between the components of high-dimensional vector data enables a lower-dimensional representation without significant loss of information. Recognizing the limitations of principal component analysis (PCA), researchers in the statistics and neural network

communities have developed nonlinear extensions of PCA. This article develops a local linear approach to dimension reduction that provides accurate representations and is fast to compute. We exercise the algorithms on speech and image data, and compare performance with PCA and with neural network implementations of nonlinear PCA. We find that both nonlinear techniques can provide more accurate representations than PCA and show that the local linear techniques outperform neural network implementations.

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91. A Constructive, Incremental-Learning Network for Mixture Modeling and Classification

James R. Williamson

https://doi.org/10.1162/neco.1997.9.7.1517 | pp. 1517-1543

Gaussian ARTMAP (GAM) is a supervised-learning adaptive resonance theory (ART) network that uses gaussian-defined receptive fields. Like other ART networks, GAM incrementally learns and constructs a representation of sufficient complexity to solve a problem it is trained on. GAM's representation is a gaussian mixture model of the input space, with learned mappings from the mixture components to output classes. We show a close relationship between GAM and the well-known expectation-maximization (EM) approach to mixture modeling. GAM outper forms an EM classification algorithm on three classification benchmarks, thereby demonstrating the advantage of the ART match criterion for regulating learning and the ARTMAP match tracking operation for incorporating environmental feedback in supervised learning situations.

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92. Shape Quantization and Recognition with Randomized Trees

Yali Amit, Donald Geman

https://doi.org/10.1162/neco.1997.9.7.1545 | pp. 1545-1588

We explore a new approach to shape recognition based on a virtually infinite family of binary features (queries) of the image data, designed to accommodate prior information about shape invariance and regularity. Each query corresponds to a spatial arrangement of several local topographic codes (or tags), which are in themselves too primitive and common to be informative about shape. All the discriminating power derives from relative angles and distances among the tags. The important attributes of the queries are a natural partial ordering corresponding to increasing structure and complexity; semi-invariance, meaning that most shapes of a given class will answer the same way to two queries that are successive in the ordering; and stability, since the queries are not based on distinguished points and substructures. No classifier based on the full feature set can be evaluated, and it is impossible to determine a priori which arrangements are informative. Our approach is to select informative features and build tree classifiers at the same time by inductive learning. In effect, each tree provides an approximation to the full posterior where the features chosen depend on the branch that is traversed. Due to the number and nature of the queries, standard decision tree construction based on a fixed-length feature vector is not feasible. Instead we entertain only a small random sample of queries at each node, constrain their complexity to increase with tree depth, and grow multiple trees. The terminal nodes are labeled by estimates of the corresponding posterior distribution over shape classes. An image is classified by sending it down every tree and aggregating the resulting distributions. The method is applied to classifying handwritten digits and synthetic

linear and nonlinear deformations of three hundred [Formula: see text] symbols. State-of-the-art error rates are achieved on the National Institute of Standards and Technology database of digits. The principal goal of the experiments on [Formula: see text] symbols is to analyze invariance, generalization error and related issues, and a comparison with artificial neural networks methods is presented in this context. [Figure: see text]

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93. Airline Crew Scheduling with Potts Neurons

M. Lagerholm, Carsten Peterson, Bo Söderberg

https://doi.org/10.1162/neco.1997.9.7.1589 | pp. 1589-1599

A Potts feedback neural network approach for finding good solutions to resource allocation problems with a nonfixed topology is presented. As a target application, the airline crew scheduling problem is chosen. The topological complication is handled by means of a propagator defined in terms of Potts neurons. The approach is tested on artificial random problems tuned to resemble real-world conditions. Very good results are obtained for a variety of problem sizes. The computer time demand for the approach only grows like (number of flights) 3 . A realistic problem typically is solved within minutes, partly due to a prior reduction of the problem size, based on an analysis of the local arrival and departure structure at the single airports.

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94. Online Learning in Radial Basis Function Networks

Jason A. S. Freeman, David Saad

https://doi.org/10.1162/neco.1997.9.7.1601 | pp. 1601-1622

An analytic investigation of the average case learning and generalization properties of radial basis function (RBFs) networks is presented, utilizing online gradient descent as the learning rule. The analytic method employed allows both the calculation of generalization error and the examination of the internal dynamics of the network. The generalization error and internal dynamics are then used to examine the role of the learning rate and the specialization of the hidden units, which gives insight into decreasing the time required for training. The realizable and some over realizable cases are studied in detail: the phase of learning in which the hidden units are unspecialized (symmetric phase) and the phase in which asymptotic convergence occurs are analyzed, and their typical properties found. Finally, simulations are performed that strongly confirm the analytic results.

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95. Minimax Entropy Principle and Its Application to Texture Modeling

Song Chun Zhu, Ying Wu, David Mumford

https://doi.org/10.1162/neco.1997.9.8.1627 | pp. 1627-1660

This article proposes a general theory and methodology, called the minimax entropy principle, for building statistical models for images (or signals) in a variety of applications. This principle consists of two parts. The first is the maximum entropy principle for feature binding (or fusion): for a given set of observed feature statistics, a distribution can be built to bind these feature statistics together by maximizing the entropy over all distributions that reproduce them. The second part is the minimum entropy principle for feature selection: among all plausible sets of feature statistics, we choose the set whose maximum entropy distribution has the

minimum entropy. Computational and inferential issues in both parts are addressed; in particular, a feature pursuit procedure is proposed for approximately selecting the optimal set of features. The minimax entropy principle is then corrected by considering the sample variation in the observed feature statistics, and an information criterion for feature pursuit is derived. The minimax entropy principle is applied to texture modeling, where a novel Markov random field (MRF) model, called FRAME (filter, random field, and minimax entropy), is derived, and encouraging results are obtained in experiments on a variety of texture images. The relationship between our theory and the mechanisms of neural computation is also discussed.

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96. A Local Learning Rule That Enables Information Maximization for Arbitrary Input Distributions

Ralph Linsker

https://doi.org/10.1162/neco.1997.9.8.1661 | pp. 1661-1665

This note presents a local learning rule that enables a network to maximize the mutual information between input and output vectors. The network's output units may be nonlinear, and the distribution of input vectors is arbitrary. The local algorithm also serves to compute the inverse C –1 of an arbitrary square connection weight matrix.

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97. Convergence and Ordering of Kohonen's Batch Map

Yizong Cheng

https://doi.org/10.1162/neco.1997.9.8.1667 | pp. 1667-1676

The convergence and ordering of Kohonen's batch-mode self-organizing map with Heskes and Kappen's (1993) winner selection are proved. Selim and Ismail's (1984) objective function for k-means clustering is generalized in the convergence proof of the self-organizing map. It is shown that when the neighborhood relation is doubly decreasing, order in the map is preserved. An unordered map becomes ordered when a degenerate state of ordering is entered, where the number of distinct winners is one or two. One strategy to enter this state is to run the algorithm with a broad neighborhood relation.

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98. Solitary Waves of Integrate-and-Fire Neural Fields

D. Horn, Irit Opher

https://doi.org/10.1162/neco.1997.9.8.1677 | pp. 1677-1690

Arrays of interacting identical neurons can develop coherent firing patterns, such as moving stripes that have been suggested as possible explanations of hallucinatory phenomena. Other known formations include rotating spirals and expanding concentric rings. We obtain all of them using a novel two-variable description of integrate-and-fire neurons that allows for a continuum formulation of neural fields. One of these variables distinguishes between the two different states of refractoriness and depolarization and acquires topological meaning when it is turned into a field. Hence, it leads to a topologic characterization of the ensuing solitary waves, or excitons. They are limited to pointlike excitations on a line and linear excitations, including all the examples noted above, on a two-dimensional surface. A moving patch of firing activity is not an allowed solitary wave on our neural surface. Only the presence of strong inhomogeneity that destroys the neural field

continuity allows for the appearance of patchy incoherent firing patterns driven by excitatory interactions.

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99. Time-Series Segmentation Using Predictive Modular Neural Networks

Athanasios Kehagias, Vassilios Petridis

https://doi.org/10.1162/neco.1997.9.8.1691 | pp. 1691-1709

A predictive modular neural network method is applied to the problem of unsupervised time-series segmentation. The method consists of the concurrent application of two algorithms: one for source identification, the other for time-series classification. The source identification algorithm discovers the sources generating the time series, assigns data to each source, and trains one predictor for each source. The classification algorithm recursively computes a credit function for each source, based on the competition of the respective predictors, according to their predictive accuracy; the credit function is used for classification of the time-series observation at each time step. The method is tested by numerical experiments.

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100. Adaptive Mixtures of Probabilistic Transducers

Yoram Singer

https://doi.org/10.1162/neco.1997.9.8.1711 | pp. 1711-1733

We describe and analyze a mixture model for supervised learning of probabilistic transducers. We devise an online learning algorithm that efficiently infers the structure and estimates the parameters of each probabilistic transducer in the mixture. Theoretical analysis and comparative simulations indicate that the learning algorithm tracks the best transducer from an arbitrarily large (possibly infinite) pool of models. We also present an application of the model for inducing a noun phrase recognizer.

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101. Long Short-Term Memory

Sepp Hochreiter, Jürgen Schmidhuber

https://doi.org/10.1162/neco.1997.9.8.1735 | pp. 1735-1780

Learning to store information over extended time intervals by recurrent backpropagation takes a very long time, mostly because of insufficient, decaying error backflow. We briefly review Hochreiter's (1991) analysis of this problem, then address it by introducing a novel, efficient, gradient-based method called long short-term memory (LSTM). Truncating the gradient where this does not do harm, LSTM can learn to bridge minimal time lags in excess of 1000 discrete-time steps by enforcing constant error flow through constant error carousels within special units. Multiplicative gate units learn to open and close access to the constant error flow. LSTM is local in space and time; its computational complexity per time step and weight is O(1). Our experiments with artificial data involve local, distributed, real-valued, and noisy pattern representations. In comparisons with real-time recurrent learning, back propagation through time, recurrent cascade correlation, Elman nets, and neural sequence chunking, LSTM leads to many more successful runs, and learns much faster. LSTM also solves complex, artificial long-time-lag tasks that have never been solved by previous recurrent network algorithms.

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102. Factor Analysis Using Delta-Rule Wake-Sleep Learning

Radford M. Neal, Peter Dayan

https://doi.org/10.1162/neco.1997.9.8.1781 | pp. 1781-1803

We describe a linear network that models correlations between real-valued visible variables using one or more real-valued hidden variables-a factor analysis model. This model can be seen as a linear version of the Helmholtz machine, and its parameters can be learned using the wake-sleep method, in which learning of the primary generative model is assisted by a recognition model, whose role is to fill in the values of hidden variables based on the values of visible variables. The generative and recognition models are jointly learned in wake and sleep phases, using just the delta rule. This learning procedure is comparable in simplicity to Hebbian learning, which produces a somewhat different representation of correlations in terms of principal components. We argue that the simplicity of wake-sleep learning makes factor analysis a plausible alternative to Hebbian learning as a model of activity-dependent cortical plasticity.

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103. Data Clustering Using a Model Granular Magnet

Marcelo Blatt, Shai Wiseman, Eytan Domany

https://doi.org/10.1162/neco.1997.9.8.1805 | pp. 1805-1842

We present a new approach to clustering, based on the physical properties of an inhomogeneous ferromagnet. No assumption is made regarding the underlying distribution of the data. We assign a Potts spin to each data point and introduce an interaction between neighboring points, whose strength is a decreasing function of the distance between the neighbors. This magnetic system exhibits three phases. At very low temperatures, it is completely ordered; all spins are aligned. At very high temperatures, the system does not exhibit any ordering, and in an intermediate regime, clusters of relatively strongly coupled spins become ordered, whereas different clusters remain uncorrelated. This intermediate phase is identified by a jump in the order parameters. The spin-spin correlation function is used to partition the spins and the corresponding data points into clusters. We demonstrate on three synthetic and three real data sets how the method works. Detailed comparison to the performance of other techniques clearly indicates the relative success of our method.